



ALERT FOR ALL

Grade 08

ALIGNED WITH INTERNATIONAL STANDARDS

This book supports essential educational frameworks that shape future-ready minds:

- 21st Century Skills
- Sustainable Development Goals (SDGs)
- Social-Emotional Learning (SEL)
- Universal Design for Learning (UDL)

STEAM Scouts is a ground-breaking STEAM book series with an interdisciplinary focus designed to ignite curiosity, creativity, and critical thinking in learners worldwide. The series supports Social-Emotional Learning (SEL) by promoting core competencies like self-awareness, self-management, social awareness, relationship skills, and responsible decision-making. It also aligns with the UN Sustainable Development Goals (SDGs), fosters 21st Century Skills, and meets international educational standards to equip learners with tools for a dynamic, innovative, and impact-driven future. Furthermore, it is built on Universal Design for Learning (UDL) principles, ensuring inclusive, accessible, and engaging STEAM education for all students, regardless of learning ability.

Key Features

Standards-Aligned

- NGSS (Next Generation Science Standards) – Authentic scientific inquiry & engineering design.
- CCSS (Common Core State Standards) – Rigorous literacy & math integration.
- Cambridge & IB (PYP/MYP/DP) – Global perspectives & interdisciplinary learning.
- ISTE (Technology Standards) – Digital literacy & computational thinking.
- STEL (Standards for Technological & Engineering Literacy) – Real-world problem-solving.

Social-Emotional Learning (SEL) and Global Citizenship

- Each book fosters collaboration, resilience, leadership, and ethical decision-making while connecting to multiple UN Sustainable Development Goals (SDGs) - such as Quality Education (SDG 4), Clean Energy (SDG 7), Climate Action (SDG 13), and Innovation (SDG 9) - empowering students to think globally, encourages ethical innovation & community impact.
- Differentiated challenges to meet diverse learning needs and skill levels.
- Visual cues and infographics to enhance understanding and engagement.
- Guided instruction to support learners through new or complex concepts.
- Scaffolded activities that build skills progressively and promote learner independence.

Why Choose This Series?

Future-Ready Skills – Prepares learners with essential skills like creativity, collaboration, and critical thinking for future careers and global challenges.

Teacher & Student-Friendly – Includes engaging lesson hooks, discussion prompts, guided instruction, and hands-on tasks for seamless learning.

- **SDG-Themed Modules** – Each module links to 2+ UN SDGs, blending academics with global purpose and ethical action.
- **Inclusive & Engaging** – Built on UDL principles to ensure accessibility through differentiated pathways, visual supports, and hands-on learning.

Perfect For:

- K-12 Schools adopting IB, Cambridge, NGSS, or CCSS
- STEM/STEAM Coordinators seeking industry-relevant curricula
- Homeschoolers and Educators wanting SDG-aligned, future-focused content
- Educational Institutions committed to sustainability and inclusion

Theme

03

ALERT FOR ALL

Each week has a double session, total time of session 80-90 mins



Week	Unit Name	Lesson	Page
Week 1	Unit 1: Spark The Challenge!	Lesson 1: Sensing danger- Saving lives.	2-18
Week 2		Lesson 2: Danger in the Air!	19-36
Week 3		Lesson 3: Engineers in Action!	37-52
Week 4	Unit 2: Innovators in Action.	Lesson 1: The Brain Behind the System.	53-71
Week 5		Lesson 2: Code Red!	72-93
Week 6		Lesson3: A Safer World.	94-112
Week 7	Unit 3: Beyond Imagination.	Lesson 1: Into the Emergency Simulator.	113-135
Week 8		Lesson 2: Connected & protected.	136-153

About The Book

Everyone deserves to feel safe—regardless of where they are or what their abilities may be. Yet in emergencies like gas leaks, fires, or extreme heat, not everyone receives timely, accessible warnings. What if technology could detect danger, respond instantly, and deliver alerts in ways that work for everyone? In this theme, students become young inventors, engineers, and coders as they design a Multi-Sensory Smart Alert System that uses light, sound, vibration, and mobile technology to protect lives and promote safety.

Students will explore how real-world sensors—such as the MQ-2 gas sensor and the DHT11 temperature/humidity sensor—can monitor environmental changes. With Arduino microcontrollers, they'll build responsive systems that trigger multiple alert types (LEDs, buzzers, vibrating motors, and digital messages) to accommodate a range of user needs, including those who are deaf, blind, or have sensory sensitivities.

Taking the challenge a step further, students will integrate AI concepts like pattern recognition and threshold-based decision making, enabling their system to identify specific danger conditions and respond intelligently. They'll also develop mobile apps using MIT App Inventor and Bluetooth modules, displaying real-time sensor data, sending push notifications, and allowing remote control of alert settings—an introduction to IoT (Internet of Things) systems and AI-assisted responsiveness.

Throughout this experience, students will deepen their understanding of:

- Sensor-based input and automated output systems
- Block coding, conditionals, and event-driven programming
- Wireless communication and mobile UI design
- Inclusive, human-centered engineering
- Real-time data processing and decision-making logic

This project aligns with Sustainable Development Goals 3 (Good Health and Well-being), 9 (Industry, Innovation, and Infrastructure), and 10 (Reduced Inequalities). By the end of the theme, students will have built a fully functional, multi-sensory alert system—and discovered how AI, coding, and empathy can work together to create safer, smarter, and more inclusive communities for all.

HOW TO USE THIS BOOK?

1

Ready, Set, Learn!

Where we merge your existing knowledge with exciting new discoveries.

2

Inquiry-Based Learning!

Let's Think!

In this section, students will brainstorm and explore possible solutions to specific problems. This encourages critical thinking and creativity as they consider various approaches.

Explorer's Toolkit

Here, students will dive deeper into the topic. They will engage with curated videos, articles, educational games, and hands-on experiments to broaden their understanding and spark curiosity.

DIY Lab

It's part of the explore, students will undertake hands-on projects where they will design, build, and test their own creations. They will apply scientific principles, use technology and engineering skills, explore artistic concepts, and utilize mathematical reasoning.

Screen Lessons

Students will watch videos that explain key concepts in an engaging and accessible way.

3

Problem-Based Learning:

- **Introducing the problem:**
Students will join a journey to tackle real-world environmental challenges. Through hands-on activities and engaging tasks, they will apply their knowledge and skills to discover innovative solutions.
- **Apply the Engineering Design Process (EDP):**
Students will use a structured approach to define problems, conduct research, generate ideas, prototype solutions, test, evaluate, and refine their designs.
- **The Scientific Method:**
Students will follow a systematic process of inquiry—making observations, forming hypotheses, conducting experiments, analyzing data, and drawing conclusions—to develop a deeper understanding of scientific concepts and problem-solving. In doing so, they will incorporate the CER framework to present their scientific claims, provide evidence from their experiments, and explain their reasoning behind the conclusions they draw.

4

Project-Based Learning:

PBL immerses students in real-world challenges through hands-on projects. They build critical thinking, creativity, collaboration, and communication skills while designing solutions and taking ownership of their learning.

5

Story

Join Adam and Laila, two fearless STEAM Scouts, on adventures full of discovery and creativity! From building inventions to solving real-world challenges, they'll inspire you to think, experiment, and innovate. Step into the world of science, technology, engineering, arts, and math—where every challenge becomes an adventure!

Story Reflection

In this section, students will reflect on the story's key themes, characters, and challenges.

6

Assessment Exercises:

These are divided into three categories to ensure comprehensive learning and mastery.

Apprentice:

Developing understanding and can apply skills with guidance.

Builder:

Demonstrates solid understanding and can apply skills independently.

Mastermind:

Demonstrates deep understanding and can apply skills creatively and strategically.

7

Career Readiness

This section equips students with essential skills for future careers by fostering problem-solving, teamwork, and adaptability. Through hands-on projects, real-world applications, and critical thinking challenges, students develop the confidence and competencies needed for success in the ever-evolving job market.

8

Mission Innovate

Now, it's your turn! In Mission Innovate; students will apply their creativity, problem-solving skills, and knowledge to develop their own competition projects. This challenge encourages independent thinking, collaboration, and real-world innovation, empowering students to become the next generation of creators and problem-solvers.

Now I Can

This section is designed to help you measure your progress and achievements. Enjoy your exploration and learning journey through this STEAM book!

My Journal

In this part students will write their own story while they are creating the project.

By the end of this theme, students will be able to:

- Describe how human senses work together to detect danger and how impairments affect emergency awareness.
- Analyse how the human body responds to threats using scientific understanding and real-world data.
- Explain the experiences and challenges faced by individuals with hearing or vision impairments during emergencies.
- Model chemical reactions, heat transfer, and gas dispersion to understand how emergencies like fires and leaks occur.
- Apply scientific notation and unit conversions to interpret hazard-related data.
- Use the Engineering Design Process to design and prototype inclusive emergency alert solutions.
- Collect and interpret data from environmental sensors (e.g., MQ-2 for gas, DHT11 for temperature).
- Graph sensor readings and determine when an environment becomes unsafe.
- Build and program a working alert system using Arduino, LEDs, and buzzers or vibration motors.
- Develop logic-based code that varies alert intensity based on sensor input.
- Design safe evacuation routes inside a home based on sensor readings and environmental data
- Design 3D models of a room and incorporate evacuation routes into the layout.
- Build a mobile emergency app using MIT App Inventor that sends SMS alerts and responds to sensor triggers.
- Design a user-friendly app interface with high-contrast visuals, large buttons, and simple navigation.

Unit

1

Lesson

1

Spark The Challenge!

Sensing danger- Saving lives.

Time Frame: [90 min]

Lesson Objectives

By the end of this lesson students will be able to:

- Identify situations or environmental cues that signal potential danger.
- Identify the five human senses and explain their roles in detecting danger (e.g., smelling gas, hearing alarms).

Challenge questions of the lesson

- When designing an emergency alert system, what is the most important reason to use more than one type of sensory warning (like sight, sound, and touch)?

SEL Relation

1. Self-Awareness

- **Connection:** The activity requires students to reflect on their personal environment and their own body's responses. By identifying what they would see, hear, or smell as a warning sign, and how their body would react, students gain a deeper understanding of their own sensory perceptions and instinctual responses to danger.

2. Social Awareness

- **Connection:** By analyzing dangers in their home or neighborhood, students are implicitly considering the safety of the people around them—family members, friends, and the community. This encourages them to think beyond their own safety and consider the well-being of others in a shared environment.

3. Responsible Decision-Making

- **Connection:** This is the core competency of the activity. Students are tasked with not just identifying a problem but proposing a solution. This process requires them to think critically about the potential consequences of a danger and to make a responsible and well-thought-out suggestion for how to prevent it or make it safer.

4. Relationship Skills

- **Connection:** The activity encourages students to think about how their proposed safety suggestions might be communicated or implemented with others. A solution often requires collaboration, clear communication of a safety plan, or the consideration of a solution that protects everyone in a household or community.

Engage



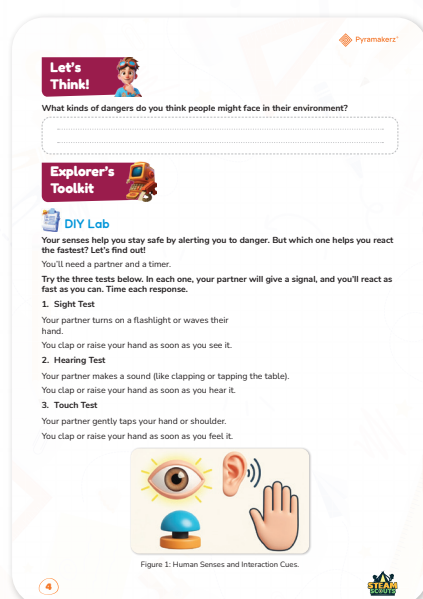
Teacher's Role:

- **Facilitate Inquiry:** Act as a guide, not a provider of answers. Provide students with access to resources, data, and case studies to help them investigate the driving questions.
- **Cultivate Empathy:** Lead discussions that encourage students to think about different sensory experiences. Connect students with community experts or online resources to help them better understand the challenges people with sensory differences face.
- **Structure the Design Process:** Help students break down the broad "Mission" into manageable steps, from initial research and ideation to prototyping and a final proposal.
- **Guide Critical Analysis:** Prompt students to justify their proposals with evidence from their research. Challenge them to consider potential limitations and unintended consequences of their solutions.

Expected from students:

- **Demonstrate Empathy:** Students will conduct research and engage in discussions to understand the perspectives and challenges of people with different sensory needs.
- **Engage in Research & Analysis:** Students will actively investigate the driving questions by collecting and analyzing data on human perception, threat dynamics, and real-world incidents.
- **Apply Creative Problem-Solving:** Students will use their insights to propose a creative and well-justified solution that addresses a specific need in accessible safety.
- **Communicate Effectively:** Students will be able to clearly articulate their findings, their proposed solution, and the rationale behind their design choices.

Parts of the book included.



Case Study



Read the Case Study Together

- Display and read the Project Overview aloud with expression.
- Pause at the line “This is a moment for problem-solvers, inventors, and change makers...”
- Ask: “What does it mean to step into the role of a problem-solver for people with different needs?” to introduce the idea of empathy.

Introduce the Mission

- Read Your Mission aloud.
- Ask students to brainstorm one challenge a person with hearing, vision, or combined sensory impairments might face during an emergency.
- Note ideas on the board.

Guide Inquiry with Driving Questions

- Display the Driving Questions list.
- Choose 1–2 questions to discuss as a whole group. For example:
 - “What does accessible safety look like?”
 - “How can we design systems that detect danger before humans do?”
- Encourage students to think beyond visible or audible cues—include vibration, visual symbols, and environmental changes.

Set the Challenge

- Explain that over the coming lessons, they will:
 1. Identify a specific problem faced by people with sensory differences in emergencies.
 2. Use empathy mapping, research, and observation.
 3. Brainstorm, design, and prototype a creative alert or evacuation system.
- Emphasize: “We’re not just learning about technology — we’re designing it to protect lives.”

Possible Strategies for Implementation:

Hook & Empathy-Building Activity

Strategy: Begin with an experiential hook to help students connect emotionally.

Implementation:

Show a short video or case study of a person with sensory impairments responding to danger.

- Lead a discussion: “What warning signs did they miss? How did it affect their safety?”

- Have students do a quick sensory simulation (blindfolds, earplugs) to feel the challenges.

Perspective-Taking Activity

Strategy: Sensory Impairment Simulation

Implementation:

- Students try completing a simple task while blindfolded or with sound muted.
- Discuss:
 - “How did that feel?”
 - “What would have helped you feel safe?”

Role-Playing Interviews

Strategy: User-Centered Design Thinking

Implementation:

- Pair students for mock interviews—one plays the role of a person with a sensory impairment, the other as an engineer.
- Provide prompt cards with possible challenges (e.g., gas leak detection, evacuation in smoke).
- Reflect: “What new ideas came from this conversation?”

Gallery Walk: Emergency Challenges

Strategy: Collaborative Idea Sharing

Implementation:

- Place posters or stations labeled Gas Leak, Fire, Extreme Heat, Silent Emergencies.
- Students post sticky notes with challenges or possible solutions.
- Vote on which type of emergency to focus on for their design.

Driving Question Launch Pad

- Display key questions (e.g., “How can we alert someone who can’t hear or see about danger?”).
- Create a digital question wall for brainstorming.
- Break into small groups to share personal observations or real-world examples.

Possible Class Discussion Points:

- Sensory limitations can make emergencies more dangerous.
- Technology can detect threats faster than humans.
- Safety should include visual, audio, and tactile alerts.

Let's Think!



Objective:

To activate students' prior knowledge and prompt them to think critically about the various types of dangers and threats present in different environments. This serves as a foundational step for a larger project on accessible safety.

Answers:

- **Natural Disasters:** Fires, floods, earthquakes, hurricanes, tornadoes, extreme temperatures.
- **Household Dangers:** Gas leaks, electrical fires, carbon monoxide leaks, chemical spills, falls, and other physical hazards.
- **Environmental Threats:** Air and water pollution, toxins, hazardous waste.
- **Man-Made/Situational Dangers:** Car accidents, security threats, loud noises, and other emergencies in public spaces.
- **Health and Safety Risks:** Food allergies, medical emergencies, infectious diseases.

Possible Strategies for Implementation:

1. Brainstorming Web or Mind Map

- **How to Implement:** Write the central question, "What kinds of dangers do you think people might face in their environment?" on a whiteboard or large paper. As students call out ideas, draw a line from the central question to their response. Encourage students to build on each other's ideas, creating a visual web of interconnected threats. This is a great way to show the breadth of the topic.

2. Think-Pair-Share

- **How to Implement:** First, ask students to spend 2-3 minutes thinking about the question and writing down their own ideas individually. Then, have them turn to a partner and share their thoughts, adding their partner's ideas to their list. Finally, facilitate a class discussion where pairs share their combined list, ensuring every student has a chance to contribute.

3. Categorization Activity

- **How to Implement:** After an initial brainstorming session, write the various student-generated dangers on the board. Then, work with the class to group the items into logical categories (e.g., "Natural," "Man-Made," "Chemical," "Physical"). This helps students move from a simple list to a more conceptual understanding of different types of risks.

Explorer's Toolkit



Teacher's Role:

- **Instructions & Safety:** Introduce the activity, explaining its purpose and providing clear instructions. Ensure students are conducting the experiment safely and fairly with their partners.
- **Oversight & Data Quality:** Circulate around the room to monitor the partner work, ensuring students are using the timer correctly and accurately recording their data.
- **Facilitator of Discussion:** Lead the "Reflect and Discuss" portion of the activity, guiding students to connect their personal findings to the larger scientific principles of sensory perception and the project's theme of accessible safety.

Expected From Students:

- **Collaborative Work:** Students will work cooperatively with their partners to conduct the three reaction time tests.
- **Accurate Data Collection:** Students will use a timer correctly to measure their reaction times and record the data in the provided chart.
- **Critical Reflection:** Students will participate in a class discussion, reflecting on their data and answering the guiding questions to connect their findings to real-world scenarios.

Parts of the book included.

Pyramakerz®

Let's Think!

What kinds of dangers do you think people might face in their environment?

Explorer's Toolkit

DIY Lab

Your senses help you stay safe by alerting you to danger. But which one helps you react the fastest? Let's find out!

You'll need a partner and a timer.

Try the three tests below. In each one, your partner will give a signal, and you'll react as fast as you can. Time each response.

- 1. Sight Test**
Your partner turns on a flashlight or waves their hand.
You clap or raise your hand as soon as you see it.
- 2. Hearing Test**
Your partner makes a sound (like clapping or tapping the table).
You clap or raise your hand as soon as you hear it.
- 3. Touch Test**
Your partner gently taps your hand or shoulder.
You clap or raise your hand as soon as you feel it.

Figure 1: Human Senses and Interaction Cues.

4

Pyramakerz®

Record and compare.

Sense Tested	Reaction Time (seconds)
Sight	
Hearing	
Touch	

Reflect and Discuss

- Which sense helped you react the fastest?
- Was there a big difference between the three?
- Why do you think our bodies respond to danger in different ways?
- How do our senses help protect us in real emergencies?

Screen Lessons

Let's watch this video to learn how our senses respond to any stimuli and danger:

<https://www.youtube.com/watch?v=fMrz3L9bMsl&t=212s>

5



Objective:

To conduct an experiment to measure and compare reaction times to visual, auditory, and tactile stimuli. Students will analyze their data to empirically understand how different senses contribute to threat detection and response.

Possible answers:

Which sense helped you react the fastest?

- Possible answer: Touch was the fastest because the signal goes directly from the skin's nerve endings to the brain with less processing.
- Possible answer: Hearing was fastest because sound signals reach the brain quickly.

Was there a big difference between the three?

- Possible answer: Yes, the reaction times for touch and hearing were noticeably faster than sight.
- Possible answer: No, all three senses had similar reaction times.

Why do you think our bodies respond to danger in different ways?

- Possible answer: Each sense processes information differently—sight requires the brain to interpret visual information, while touch and hearing often trigger quicker reflexes.
- Possible answer: Some senses are more direct in sending danger signals to the brain.

How do our senses help protect us in real emergencies?

- Possible answer: They alert us to danger before we can see it—like feeling heat before seeing flames or hearing an alarm before seeing smoke.
- Possible answer: They give us multiple ways to detect threats, so if one sense is impaired, others can still warn us.

Materials Needed:

- A timer (phone or classroom clock).
- Common classroom objects that can make a distinct sound (e.g., dropping a pencil, clapping, clicking fingers - for "Hearing (sound)").
- A light source (e.g., flashlight, classroom lights - for "Light (sight)").

Activity Instructions:

- **Find a Partner:** You will work in a pair for this experiment.
- **Gather Materials:** Collect a timer (or smartphone with a stopwatch app) and a pencil/pen and a data sheet to record your results.
- **Decide Roles:** Choose who will be the "Reactor" and who will be the "Signaler and Timer" for the first round of tests. You will switch roles later.
- **Prepare for the Sight Test:** The Reactor should get ready to react, and the Signaler should get ready to give a visual signal (a hand wave or flashing a light) and start the timer at the exact same moment.

- **Conduct the Sight Test:** The Signaler gives the signal and starts the timer. The Reactor reacts as fast as possible by clapping or raising their hand. The Signaler stops the timer and records the reaction time in the chart.
- **Conduct the Hearing Test:** Repeat the process, but this time the Signaler gives an auditory signal (a clap or table tap) and starts the timer.
- **Conduct the Touch Test:** Repeat the process again. The Signaler gives a tactile signal (a gentle tap on the shoulder or hand) and starts the timer.
- **Record Data:** Write down the reaction times for all three tests in the designated chart on your data sheet.
- **Analyze and Discuss:** With your partner, look at your recorded times. Discuss which sense produced the fastest reaction and reflect on the questions provided.
- **Switch Roles:** Now, switch roles and repeat the entire process so both partners get to experience all three tests.

Possible Strategies for Implementation:

1. Modeling the Experiment

- **How to Implement:** Before starting, the teacher should model each of the three tests with a student volunteer. This ensures that every student understands the procedure for giving a signal, starting the timer, and reacting.

2. Data Aggregation and Analysis

- **How to Implement:** After pairs have completed their tests, create a class chart on the board. Ask pairs to share their data for each test. Calculate and discuss the class average for sight, hearing, and touch. This provides a larger sample size for discussion and can lead to a more statistically robust conclusion.

3. Structured Discussion

- **How to Implement:** Use the "Reflect and Discuss" questions as a script for a guided conversation. Start with the data-driven questions ("Which sense was fastest?") and then transition to the more conceptual questions ("Why do you think there's a difference?"). This helps scaffold the discussion from concrete findings to abstract concepts.

4. Challenge Extension

- **How to Implement:** For a quick extension, challenge students to come up with and test a fourth sense (e.g., a "smell" test with a common odor like hand sanitizer, or a "taste" test with a safe food item). This encourages students to think creatively about the limits of their sensory systems.

Explain



Teacher's Role:

- Introduce the purpose of watching the video and connect it to the activity.
- Guide students to actively engage with the video by asking reflective questions before and during viewing.
- Facilitate discussions and reflections after watching to ensure understanding of key concepts.
- Support vocabulary development by explaining key terms before the video.

Expected From Students:

- Engage actively by listening and observing during the video.
- Respond to teacher prompts and questions during pauses.
- Share their observations and insights during post-viewing discussions.
- Participate in creative or reflective activities after the video to solidify their understanding.

Parts of the book included.



Screen Lessons

Let's watch this video to learn how our senses respond to any stimuli and danger:

<https://www.youtube.com/watch?v=fMrz3L9bMsl&t=212s>



Screen Lessons

Objective:

Students will watch an educational video about the brain and senses to understand how different parts of the brain control sensory functions and react to danger. They will connect brain functions to sensor technology and reflect on the design of assistive technologies.

Link: <https://www.youtube.com/watch?v=fMrz3L9bMsl&t=212s>



Possible Strategies for Implementation:

1. Pre-Watching: Set a Purpose

Strategy: K-W-L Chart

- Prompt 1: "What do you already know about the brain and senses?"

Expected Answers:

- "The brain controls how we think and move."
- "We have five senses—sight, sound, smell, touch, and taste."

- “The brain receives messages from the body.”
- Prompt 2: “What do you want to learn from this video?”

Expected Answers:

- “How the brain understands what we feel or hear.”
- “Which parts of the brain control different senses?”
- “How fast the brain reacts to danger.”

2. During Watching: Active Engagement

Strategy: Pause & Prompt (at ~1:30, ~3:30, ~6:00)

- Prompt 1: “What part of the brain was discussed?”

Expected Answers:

- “The cerebrum for thinking and senses.”
- “The cerebellum for balance and coordination.”
- “The brainstem for automatic responses.”
- Prompt 2: “What sense does it control?”

Expected Answers:

- “The cerebrum helps process sight and sound.”
- “The cerebellum helps with movement and touch.”
- “The brainstem helps us breathe and respond to pain.”
- Prompt 3: “How is this like a sensor in technology?”

Expected Answers:

- “Sensors also collect information and send it to a system to respond.”
- “Like a sensor detects heat, our skin senses temperature.”
- “Both react to changes around them.”

3. Post-Watching: Discussion & Reflection

Strategy: Think–Pair–Share + Complete K-W-L Chart

- Prompt 1: “What did you find most interesting?”

Expected Answers:

- “How the brain sends messages super-fast.”
- “How different parts of the brain do different jobs.”
- Prompt 2: “How is your brain like a sensor system?”

Expected Answers:

- “It collects signals and decides what to do, like a processor.”
- “It sends alerts to your body, like an alarm system.”
- Prompt 3: “How does this help us design better assistive tech?”

Expected Answers:

- “We can use sensors like eyes and ears in machines.”
- “We can create tools that help people who can’t hear or see.”
- “It helps us know we need to include different types of alerts.”

Elaborate and Evaluate



Teacher's Role:

- Encourage creativity and application of knowledge facilitate the assessment
- provide guidance for the Builder activity.
- Encourage creativity and application of knowledge.
- Guide students in their design process, providing feedback and support as needed.
- Facilitate the transition from theoretical understanding to practical ideation.

Expected From Students:

- **Demonstrate Content Knowledge:** Students will answer the provided multiple-choice questions, showing an understanding of sensory processing, neural pathways, and the role of multi-sensory integration.
- **Collaborative Experimentation:** Students will work in pairs to perform the ruler drop test, accurately recording data for each trial and calculating their average reaction distance.
- **Data Analysis and Reflection:** Students will analyze their own data to answer the final reflection question, articulating what their reflex speed says about their body's quick response to an unexpected event.

Parts of the book included.

Assessment

Apprentice

1. What does sensory processing help the brain do?

A. Improve muscle strength
B. Increase blood flow to the brain
C. Understand and respond to information from the environment
D. Keep the heart beating steadily
E. Control balance only

2. What is the correct path for sensory information to travel?

A. Brain → Spinal cord → Muscles
B. Muscle → Brain → Skin
C. Sensory organ → Nerve pathway → Brain cortex
D. Brain → Eye → Nose
E. Spine → Brain → Ear

3. What does it mean when the brain integrates information from more than one sense?

A. The brain ignores one sense and focuses on the other
B. Each sense is processed in a different brain
C. The brain combines signals to form a full understanding
D. The brain makes the signals louder
E. It makes you forget what you saw or heard

4. What can happen when the brain receives confusing or conflicting sensory signals?

A. You feel very calm and focused
B. You see better
C. It can be hard to understand what's happening around you
D. The brain speeds up your reaction
E. You stop using that sense

5. How can this video help us design better emergency alert systems?

A. It explains how to program sensors
B. It shows the importance of using multiple senses to send alerts
C. It teaches how to escape danger faster
D. It shows how to build robots
E. It explains how to predict accidents

Builder

Catch It If You Can!

Reflexes help protect you by allowing your body to react instantly—sometimes even before your brain fully processes what's happening. Let's test how fast your reflexes are!

You'll need a partner and a 30 cm ruler.

Try the test below. Your partner will drop the ruler without warning, and you'll try to catch it as fast as you can.

☐ Get Ready

Your partner holds the ruler vertically, with the 0 cm mark just above your open fingers (thumb and index finger ready to catch).

☐ Catch the Ruler

Without warning you, your partner drops the ruler.

You catch it as quickly as you can.

☐ Check the Number

Look at the number on the ruler where you caught it—that tells you how far it fell before you reacted.

☐ Repeat & Record

Try the test 3 to 5 times and record the distance each time. Then find your average reaction distance.

Reaction Time Chart

Distance (cm)	Reaction Time (sec)
5 cm	
10 cm	
15 cm	
20 cm	
25 cm	
30 cm	

What does your reaction time say about how quickly your body responds to danger?

Hey friends let's wrap up!

Everyday life is full of hidden dangers—like gas leaks, fires, overheating, or even slippery floors. These dangers aren't always easy to see, so it's important to know how to spot the warning signs around us: smoke, strange smells, flashing lights, sudden sounds, or heat.

How do our senses respond to danger?

Our five senses (sight, hearing, smell, touch, and taste) work together to detect danger and send quick signals to the brain.

For example:

Eyes see smoke or flames
Nose smells gas or burning
Skin feels heat
Ears hear alarms or loud crashes

These signals help us understand what's happening and decide how to react.

What is a reflex action?

A reflex is an automatic, fast action your body makes without thinking—like pulling your hand away from something hot or blinking when something comes close to your eyes. Reflexes help keep us safe because they're faster than normal reactions. Your nervous system handles it instantly, without needing a decision from your brain.

Mastermind

Look around your home or neighbourhood and identify three situations or places where danger might occur.

For each one:

Figure 3: Identify Local Dangers.



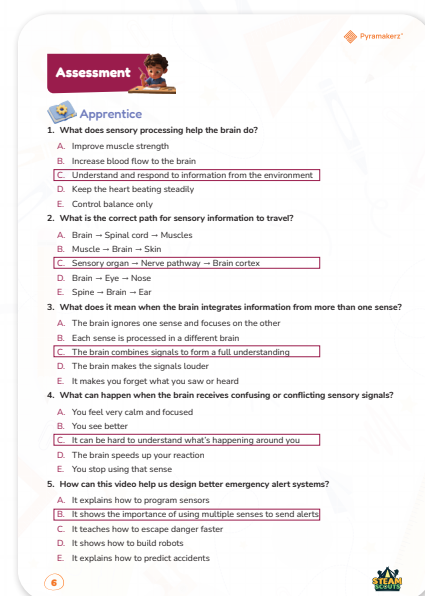
Apprentice

Objective:

To assess student comprehension of the core concepts related to sensory processing, including the path of sensory information to the brain, the role of integration, and the application of these principles to real-world safety systems.

Answers:

1. C. Understand and respond to information from the environment
2. C. Sensory organ → Nerve pathway → Brain cortex
3. C. The brain combines signals to form a full understanding
4. C. It can be hard to understand what's happening around you
5. B. It shows the importance of using multiple senses to send alerts



Possible Strategies for Implementation:

1. **Quick Check-in:** Use these questions as a quick, no-stakes quiz immediately after a video or a lesson on sensory processing. This serves as a formative assessment to identify which students have grasped the key concepts.
2. **Think-Pair-Share:** Have students first answer the questions individually. Then, in pairs, they can discuss their answers and try to come to a consensus on the correct choice. Finally, the teacher can lead a class-wide discussion to review each question and clarify any misconceptions.
3. **Individual Whiteboard Response:** Provide each student with a small whiteboard. Read each question aloud and have students write their answer (A, B, C, etc.) on their board. On the count of three, they all hold up their whiteboards, allowing the teacher to get an instant visual check of the class's understanding.
4. **Targeted Review:** After the questions are answered, go through each one and explain not only the correct answer but also why the incorrect options are wrong. This reinforces the learning and deepens students' understanding of the concepts.



Builder

Objective:

To conduct a ruler drop test to measure and analyze personal reaction time, and to understand the concept of a reflex as a quick, instinctual response to an unexpected stimulus.

Materials Needed:

- A partner
- A 30 cm ruler
- A pencil or pen
- A data sheet or notebook to record results

Activity Instructions:

Catch It If You Can! Reflexes help protect you by allowing your body to react instantly—sometimes even before your brain fully processes what’s happening. Let’s test how fast your reflexes are!

You’ll need a partner and a 30 cm ruler. Try the test below. Your partner will drop the ruler without warning, and you’ll try to catch it as fast as you can.

- **Get Ready**
 - Your partner holds the ruler vertically, with the 0 cm mark just above your open fingers (thumb and index finger ready to catch).
- **Catch the Ruler**
 - Without warning you, your partner drops the ruler.
 - You catch it as quickly as you can.
- **Check the Number**
 - Look at the number on the ruler where you caught it—that tells you how far it fell before you reacted.
- **Repeat & Record**
 - Try the test 3 to 5 times and record the distance each time. Then find your average reaction distance.
- **Reflect and Discuss**
 - What does your reaction time say about how quickly your body responds to danger?

Possible Strategies for Implementation:

1. Pair & Role Switch

- Students work in pairs, taking turns being the "dropper" and the "catcher."
- This ensures both students practice observation and reflex testing.

2. Demonstrate First

- Teacher models the activity in front of the class, showing correct hand position, how to drop the ruler, and how to read the measurement.
- Emphasize the “no warning” rule to ensure genuine reaction time.

3. Use Visual & Verbal Cues

- Encourage the dropper to vary the timing of the drop so the catcher can’t predict it.
- Remind students to stay focused and keep their eyes on the ruler at all times.

4. Multiple Trials & Averages

- Require each student to perform at least 3–5 trials.
- Guide them to record each result and calculate the average reaction distance for more accurate data.

5. Connect to Real-Life Examples

- After recording results, discuss how reflex speed can make a difference in real emergencies (e.g., avoiding a fall, stopping while driving, or reacting to a sudden hazard).

6. Data Visualization Extension

- Students can graph their results to compare reaction times among classmates.
- This could be integrated into a math lesson on averages, bar graphs, or scatter plots.



Mastermind

Objective:

To apply concepts of sensory perception, threat detection, and body response to analyze real-world dangers, and to propose solutions for improving safety and prevention in a student's own environment.

Activity Instructions:

- **Identify a Danger:** Think of your home or neighborhood and identify a specific situation or place where a danger might occur. For example, a gas stove, a busy street corner, or a ladder in the garage.
- **Describe the Threat:** On your worksheet or in your notebook, write a clear description of what the danger is and what could happen.
- **Find the Warning Signs:** Consider each of your senses and write down what sensory warnings you would receive if that danger were to happen (e.g., the smell of gas, the sound of a fire alarm, or the sight of a car driving too fast).
- **Describe Your Response:** Think about how your body would react to those warning signs. Would you have a sudden reflex? Would you consciously move away or call for help? Write down your expected response.
- **Propose a Solution:** Brainstorm and write down one or more ways this danger could be prevented or made safer. This could be a new device, a change in behavior, or a new rule.
- **Repeat:** Repeat steps 1 through 5 for two more different situations or places.
- **Complete the Data Table:** Fill in the provided data table with a summary of your findings for all three dangers.
- **Answer the Bonus Question:** Reflect on the last time you used a reflex and write down what triggered it.



Now I Can

Objective:

Students will reflect on their learning and identify how they've grown in understanding communication, empathy, and the role of assistive tools in helping others share ideas.

Activity Instructions:

1. Introduce the Reflection

- Say: "Let's look back on everything we've explored this lesson. You learned how it feels to face communication challenges, and how tools can help people share their thoughts in special ways."

2. Guide Students Through the Checklist

- Read each statement aloud one at a time.
- After each one, ask students to give a thumb up, sideways, or down to show how confident they feel.
- Example prompts:
 - "Did you learn how people share ideas without talking?"
 - "Do you know how tools can help people talk?"

3. Student Completion

- Let students check off or color the "Now I Can" boxes they feel good about.
- Remind them: "It's okay if you didn't check all of them. We're all still learning."

4. Optional Extension

- Ask students to circle the one they're most proud of.
- Invite them to write a sentence or draw a picture to show their favorite learning moment.

Relation to Standards

1. English Language Arts (ELA)

- **Writing Standards:** The project requires students to research, analyze information, and communicate their findings. This aligns with standards for conducting research projects to answer a question, gathering relevant information from multiple sources, and writing arguments to support claims with evidence.
- **Speaking and Listening Standards:** As students collaborate on their projects and present their final designs, they are practicing standards related to engaging in collaborative discussions, presenting findings clearly and logically, and responding thoughtfully to diverse perspectives.
- **Reading Standards for Informational Text:** Students will need to read and comprehend informational texts (e.g., articles on sensory science, case studies of emergencies) to build the background knowledge necessary for their project.

2. Next Generation Science Standards (NGSS)

- **Science and Engineering Practices (SEP):** The entire project is built on these practices, specifically:
 - Asking Questions and Defining Problems: Students are tasked with defining a real-world problem (accessible safety).
 - Planning and Carrying Out Investigations: Students conduct the ruler drop test and the real-world risk analysis, collecting and interpreting data.
 - Constructing Explanations and Designing Solutions: Students use their scientific knowledge to design and propose a solution to a problem.
- **Disciplinary Core Ideas (DCI):**
 - **LS1.A:** Structure and Function: The lesson focuses on how the structure of human sensory organs and the nervous system enables the function of threat detection and response.
 - **ETS1.A:** Defining and Delimiting Engineering Problems: Students must analyze a problem to specify criteria and constraints for a solution, such as considering the needs of people with different abilities.
 - **ETS1.B:** Developing Possible Solutions: Students create and test their ideas, comparing multiple solutions to find the most effective one.

3. ISTE Standards for Students (Technology)

- **Knowledge Constructor:** Students must critically curate a variety of digital and non-digital resources to construct knowledge about dangers, sensory perception, and inclusive design.
- **Innovative Designer:** The project is a design challenge where students use a design process to identify and solve a real-world problem, creating new, useful, and empathetic solutions.
- **Creative Communicator:** Students must communicate their findings and proposed solution clearly and creatively for different audiences, using a variety of media and formats.